

Fig. 8: Bubble levels application image

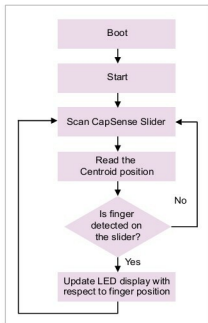


Fig. 9: CapSense flowchart

In the world of Cortex processors, operating range is up to 3.3V. A PSoC supports operating range of 1.71V to 5.5V.

A PSoC is available with programmable solutions (can be set up to implement a range of user-defined features) and configurable solutions (can implement specific features without firmware development; can be hardware through resistors or I2C register configurable) in electronic toy applications.

Increasing the number of internal and external interfaces on the board increases the number of ways that an intruder can create havoc on the system. Failure analysis and returned materials are two of the largest limitations of this embed-

ded system that can be solved by single-chip solutions.

## Design challenges

The high-performance intelligence MCU selection with higher MIPS CPU core, faster ADC ( $\geq 500\text{ksp/s}$  @ 10-bit), internal flash and SRAM memory,

internal EEPROM, analogue and digital peripherals are required to perform key functions like high-performance analogue measurement, CapSense, touchscreen interface, LCD drives, low-power operations, RTC and interfaces with different external protocols.

A PSoC supports CapSense technology. Using PSoC in the design, CapSense based button, slider and proximity can be implemented for the front panel. Meeting CapSense performance (signal-to-noise ratio) with nearby LEDs (PWM based) on the front panel is a design challenge for the system designer.

Implementing waterproofing and water-tolerance features, implementing auto-tuning with variation in trace capacitance, variation in CapSense button and slider sizes and shape are some other design challenges for the system designer, along with implementing CapSense sensing with thicker glass material (display glass) and meeting CapSense sensitivity with that type of materials.

As this solution involves electro-mechanical construction, designing a compact and cost-effective electro-mechanical solution becomes a challenge for the system designer, added with the challenge of certifying this electromechanical design with EMI/EMC standards.

Fault-detection and recovery mechanisms are required in electronic toy applications. Power supply design with battery protection, over-current, over-heating, start-up

fail condition is required for mobile phone applications. Implementing self diagnostics in electronic toy applications is also a design challenge. An MCU with one-time programmable feature prevents reverse engineering of the firmware by competitors and hackers.

## Designing electronic toys

Electronic toys can be used for raster displays, bubble levels and CapSense applications. For example, a PSoC based board can be used for showing raster display, bubble level and CapSense applications.

Components indicated in Fig. 4 are as follows:

1. PSoC 3 100-pin TQFP (Cypress CY8C3866AXI-040 TQFP100)
2. CapSense proximity sensor connector (Samtec Inc., BCS-101-L-S-HE)
3. 3-axis accelerometer (Kionix, KXSC7-2050)
4. 5-segment CapSense slider
5. Eight LEDs
6. Thermistor (Murata, NCP21X-V103J03RA)
7. 12-pin wireless expansion connector
8. 28-pin external I/O connector (22 GPIO, 2 V<sub>DD</sub>, 2 GND, 2 SIO)
9. USB programming and debugging interface (Cypress CY-7C68013A-56LFXC)
10. USB connector
11. 9V DC battery holder
12. Jumper J1 Vddio power source select
13. Jumper J4 voltage regulator source select
14. Voltage regulator (Zetex Inc., ZLDO330)
15. General-purpose push-button switch

### PSoCRocks raster image using PSoC board.

The board has a 3-axis accelerometer, which detects instant acceleration of the board in motion. Instantaneous voltage at each axis of the accelerometer is directly proportional to the acceleration along that axis. In the firmware, y-axis